1C 1 COMPARING MASSES OF SUBSTANCES

SPECIFICATION REFERENCE: 1.1 1.2 1.4

LEARNING OBJECTIVES

- Understand the terms: relative atomic mass, based on the ^{12}C scale; relative molecular mass; relative formula mass; molar mass, as the mass per mole of a substance in g mol^{-1}.
- Understand how to calculate relative molecular mass and relative formula mass from relative atomic masses.
- Perform calculations using the Avogadro constant L (6.02×10^{23} mol^{-1}).

RELATIVE ATOMIC MASS (A_r)

As chemists discovered more and more elements in the nineteenth century, they began to realise that the masses of the elements were different. They could not weigh individual atoms, but they were able to use numbers to compare the masses of atoms of different elements. For this reason, they began to use the term 'relative atomic mass'.

The chemists soon realised that the element whose atoms had the smallest mass was hydrogen, so the relative atomic mass of hydrogen was fixed as 1. Atoms of silicon had double the mass of nitrogen atoms, and nitrogen atoms were 14 times heavier than hydrogen atoms. This meant that the relative atomic mass of nitrogen was 14, and that of silicon was 28. At first, mostly whole numbers were used, but eventually it was possible to find the mass of an atom to several decimal places. The Periodic Table in the Data Booklet uses 1 decimal place for lighter elements and whole numbers for heavier ones.

After the discovery of isotopes, the ^{12}C isotope of carbon was used in the definition of relative atomic mass.

A suitable definition of relative atomic mass is:

> the weighted mean (average) mass of an atom compared to $\frac{1}{12}$ of the mass of an atom of ^{12}C

It is often useful to remember this expression:

$$A_r = \frac{\text{mean mass of an atom of an element}}{\frac{1}{12} \text{ of the mass of an atom of } ^{12}C}$$

RELATIVE MOLECULAR MASS (M_r)

Relative atomic masses are used for atoms of elements. Relative molecular masses are used for molecules of both elements and compounds. They are easily calculated by adding relative atomic masses.

Table A shows values for some common elements taken from the Data Booklet.

ELEMENT	RELATIVE ATOMIC MASS
hydrogen	1.0
carbon	12.0
oxygen	16.0
sulfur	32.1
copper	63.5

table A

Note that A_r and M_r do not have units.
Here are some examples of calculations.

WORKED EXAMPLE 1

What is the relative molecular mass of carbon dioxide, CO_2?

$M_r = 12.0 + (2 \times 16.0) = 44.0$

WORKED EXAMPLE 2

What is the relative molecular mass of sulfuric acid, H_2SO_4?

$M_r = (2 \times 1.0) + 32.1 + (4 \times 16.0) = 98.1$

EXAM HINT

Make sure you use the relative atomic masses shown on the Periodic Table in the Data Booklet.

RELATIVE FORMULA MASS (M_r)

This term has the same symbol as relative molecular mass, but the 'formula' part means that it includes both molecules and ions. Worked example 3 below is slightly more complicated because of the water of crystallisation, but there is also another problem. Hydrated copper(II) sulfate is an ionic compound, so it is not a good idea to refer to its relative *molecular* mass. That is why it is called relative *formula* mass.

WORKED EXAMPLE 3

What is the relative formula mass of hydrated copper(II) sulfate, $CuSO_4.5H_2O$?

$M_r = 63.5 + 32.1 + (4 \times 16.0) + 5\{(2 \times 1.0) + 16.0\} = 249.6$

The term 'relative formula mass' should also be used for compounds with giant structures, such as sodium chloride and silicon dioxide.

MOLAR MASS (M)

Another way around the problem in Worked example 3 is to use the term **molar mass**, which is the mass per mole of any substance (molecular or ionic). Its symbol is M (not M_r) and it has the units g mol^{-1} (grams per mole). Here we have a new term (the mole) which will be fully explained in **Topic 1C.2**. For now, you can think of one mole (1 mol) of a substance as being the same quantity as the relative formula mass of the substance, with the units of grams.

So, this is the expression you can use:

$$\text{amount in mol} = \frac{\text{mass of substance in g}}{\text{molar mass in g mol}^{-1}} \text{ or } n = \frac{m}{M}$$

Table B shows examples of working out the amounts in moles of some substances using this expression.

SUBSTANCE	O_2	CH_4	H_2O	NH_4NO_3
mass in g	5.26	4.0	100	14.7
molar mass, M in g mol^{-1}	32.0	16.0	18.0	80.0
amount in mol	0.164	0.25	5.56	0.184

table B

THE AVOGADRO CONSTANT

Amedeo Avogadro (1776–1856) was an Italian chemist whose name is used in naming the **Avogadro constant**. We are introducing him here because the scaling-up factor from atoms, molecules and ions to grams is named after him.



▲ **fig A** The Italian chemist, Amedeo Avogadro

The value of the Avogadro constant is approximately 602 000 000 000 000 000 000 000 mol^{-1}. It is easier to write this number using standard form: 6.02×10^{23} mol^{-1}.

You do not need to know a definition of the Avogadro constant, and it is best to think of it as the number of particles (atoms, molecules or ions) in one mole of any substance. For example, there are:

6.02×10^{23} helium atoms in 4.0 g of He

6.02×10^{23} carbon dioxide molecules in 44.0 g of CO_2

6.02×10^{23} nitrate ions in 62.0 g of NO_3^-

CALCULATIONS USING THE AVOGADRO CONSTANT

You will need to use the value of L in the types of calculation shown here.

1 Calculate the number of particles in a given mass of a substance. Start by using the expression:

$$\text{amount in mol} = \frac{\text{mass of substance in g}}{\text{molar mass in g mol}^{-1}} \text{ or } n = \frac{m}{M}$$

then multiply the amount in mol by the Avogadro constant.

WORKED EXAMPLE 4

How many H_2O molecules are there in 1.25 g of water?

$$n = \frac{1.25}{18.0} = 0.0694 \text{ mol}$$

number of molecules = $6.02 \times 10^{23} \times 0.0694 = 4.18 \times 10^{22}$

2 Calculate the mass of a given number of particles of a substance: start by dividing the number of particles by the Avogadro constant, then multiply the result by the molar mass.

WORKED EXAMPLE 5

What is the mass of 100 million atoms of gold?

$$n = \frac{100 \times 10^6}{6.02 \times 10^{23}} = 1.66 \times 10^{-16} \text{ mol}$$

$$m = 1.66 \times 10^{-16} \times 197.0 = 3.27 \times 10^{-14} \text{ g}$$

(There are lots of atoms, but only a tiny mass.)

CHECKPOINT

1. Malachite is an important mineral with the formula $Cu_2CO_3(OH)_2$. Calculate its relative formula mass.

2. How many molecules of sugar ($C_{12}H_{22}O_{11}$) are there in a teaspoon measure (4.20 g)?

DID YOU KNOW?

The symbol L is used for the Avogadro constant (using A would be confusing because of the use of A_r for relative atomic mass). L comes from the surname of Johann Josef Loschmidt (1821–1895), an Austrian chemist who was a contemporary of Avogadro. He made many contributions to our understanding of the same area of knowledge.

SUBJECT VOCABULARY

molar mass the mass per mole of a substance; it has the symbol M and the units g mol^{-1}

Avogadro constant (L) 6.02×10^{23} mol^{-1}, the number of particles in one mole of a substance

1C 2 CALCULATIONS INVOLVING MOLES

SPECIFICATION REFERENCE 1.2

LEARNING OBJECTIVES

- Know that the mole (mol) is the unit for the amount of a substance.

WHAT IS A MOLE?

So far, we have referred to the mole and have used it in a simple form of calculation, but we have not properly explained what it is.

THE DEFINITION OF A MOLE

A **mole** is the amount of substance that contains the same number of particles as the number of carbon atoms in exactly 12 g of the ^{12}C isotope.

This definition is not easy to understand, but will be explained as you read on.



▲ **fig A** From left to right: tin (Sn), magnesium (Mg), iodine (I) and copper (Cu). Each sample contains 6.02×10^{23} atoms.

COUNTING ATOMS

As you know, atoms are very tiny particles that cannot be seen by the human eye. When we look at the sand on a beach, we are looking at billions and billions of atoms of, mostly, silicon and oxygen in the compound silicon dioxide (SiO_2). Quoting the actual numbers of atoms involved in a reaction, whether in a test tube or on an industrial scale, would involve extremely large numbers that would be very difficult to handle.

You are already familiar with the use of relative atomic masses to compare the relative masses of atoms. In a water molecule, the oxygen atom has a mass that is 16 times greater than the mass of a hydrogen atom. You know this because in the Periodic Table the relative atomic masses are H = 1.0 and O = 16.0. Note the word 'relative'. These values do not tell us the actual mass in grams of an oxygen atom or a hydrogen atom. They only tell us that an oxygen atom has a mass 16 times greater than that of a hydrogen atom.

Now consider using these numbers (16.0 and 1.0) with the familiar unit g (grams). You can more easily visualise 16.0 g of oxygen than a single atom of oxygen. Doing this is effectively scaling up on a very large scale. The number of oxygen atoms in 16.0 g of oxygen is the same as the number of hydrogen atoms in 1.0 g of hydrogen.

CALCULATIONS USING MOLES

WHAT TO REMEMBER WHEN DOING CALCULATIONS

You can use the mole to count atoms, molecules, ions, electrons and other species. So, it is important to include an exact description of the species being referred to. Consider the examples of hydrogen, oxygen and water.

One mole of water has a mass of 18.0 g, but what is the mass of one mole of hydrogen or oxygen? It depends on whether you are referring to atoms or molecules, so you need to make this clear.

Remember that the symbol n is used for the amount of substance in mol.

Consider the substances in **table A**. The masses are the same, but the amounts are different.

	1g OF OXYGEN ATOMS (O)	1g OF OXYGEN MOLECULES (O_2)	1g OF OZONE MOLECULES (O_3)
n (amount in mol)	1 ÷ 16.0 = 0.0625 mol	1 ÷ 32.0 = 0.0313 mol	1 ÷ 48.0 = 0.0208 mol

table A

You should always clearly identify the species that you are referring to if there is any possibility that there could be more than one meaning. This is not usually necessary for compounds.

It is good practice to refer to both the formula and the name. Examples include:

- the amount, in moles, of O in 9.4 g of oxygen atoms (0.59 mol)
- the amount, in moles, of O_2 in 9.4 g of oxygen molecules (0.29 mol)
- the amount, in moles, of O_3 in 9.4 g of ozone molecules (0.20 mol)
- the amount, in moles, of CO_2 in 9.4 g of carbon dioxide (0.21 mol)
- the amount, in moles, of SO_4^{2-} in 9.4 g of sulfate ions (0.098 mol).

THE EQUATION FOR CALCULATING MOLES

The equation for calculating moles is:

$$\text{amount of substance in moles} = \frac{\text{mass in grams}}{\text{molar mass}} \text{ or } n = \frac{m}{M}$$

You will use this expression in many calculations during your study of chemistry. It is often rearranged as:

$$M = \frac{m}{n} \text{ or } m = n \times M$$

WORKED EXAMPLE 1

What is the amount of substance in 6.51 g of sodium chloride?

$$n = \frac{m}{M} = \frac{6.51}{58.5} = 0.111 \text{ mol}$$

WORKED EXAMPLE 2

What is the mass of 0.263 mol of hydrogen iodide?

$$m = n \times M = 0.263 \times 127.9 = 33.6 \text{ g}$$

WORKED EXAMPLE 3

A sample of 0.284 mol of a substance has a mass of 17.8 g. What is the molar mass of the substance?

$$M = \frac{m}{n} = \frac{17.8}{0.284} = 62.7\,\text{g mol}^{-1}$$

LEARNING TIP

When using moles, always make clear what particles you are referring to: atoms, molecules, ions or electrons. It is also a good idea to state the formula.

CHECKPOINT

1. What is the amount of substance in each of the following?

(a) 8.00 g of sulfur, S

(b) 8.00 g of sulfur dioxide, SO_2

(b) 8.00 g of sulfate ions, SO_4^{2-}

2. How many particles are there of the specified substance?

(a) atoms in 2.00 g of sulfur, S

(b) molecules in 4.00 g of sulfur dioxide, SO_2

(b) ions in 8.00 g of sulfate ions, SO_4^{2-}

SKILLS ADAPTIVE LEARNING

SUBJECT VOCABULARY

mole the amount of substance that contains the same number of particles as the number of carbon atoms in exactly 12 g of ^{12}C

1C 3 CALCULATIONS USING REACTING MASSES

SPECIFICATION REFERENCE: 1.7 1.10

LEARNING OBJECTIVES

- Use chemical equations to calculate reacting masses and vice versa, using the concepts of amount of substance and molar mass.
- Determine a formula or confirm an equation by experiment, including evaluation of the data.

INTRODUCTION TO REACTING MASSES

You can use the ideas from previous topics about amounts of substance and equations to do calculations involving the masses of reactants and products in equations.

A balanced equation for a reaction shows the same number of each species (atoms, molecules, ions or electrons) on both sides of the equation. It is also balanced for the masses of each species. This means that we can make predictions about the masses of reactants, which are needed to form a specified mass or amount of a product, or the other way round.

Consider this equation used in the manufacture of ammonia:

$$N_2 + 3H_2 \rightarrow 2NH_3$$

This shows that one molecule of nitrogen reacts with three molecules of hydrogen to form two molecules of ammonia.

This statement can be made about the amounts involved:

1 mol of N_2 reacts with 3 mol of H_2 to form 2 mol of NH_3.

This statement can be made about the masses involved:

28.0 g of N_2 reacts with 6.0 g of H_2 to form 34.0 g of NH_3.

These amounts and masses can have many other values, as long as the ratio does not change.

CALCULATING REACTING MASSES FROM EQUATIONS

Using a balanced equation, predictions can be made about reacting masses.

WORKED EXAMPLE 1

The equation for a reaction is:

$$SO_3 + H_2O \rightarrow H_2SO_4$$

What mass of sulfur trioxide is needed to form 75.0 g of sulfuric acid?

Step 1: calculate the molar masses of all substances you are told about and asked about, in this case, sulfur trioxide and sulfuric acid

$M(SO_3) = 80.1\,g\,mol^{-1}$ and $M(H_2SO_4) = 98.1\,g\,mol^{-1}$

Step 2: calculate the amount of sulfuric acid

$$n = \frac{m}{M} = \frac{75.0}{98.1} = 0.765\,mol$$

Step 3: use the reaction ratio in the equation to work out the amount of sulfur trioxide needed

As the ratio is 1 : 1, the amount is the same, so $n(SO_3) = 0.765$ mol

Step 4: calculate the mass of sulfur trioxide

$m = n \times M = 0.765 \times 80.1 = 61.2\,g$

WORKED EXAMPLE 2

The equation for a reaction is:

$$2NH_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4$$

What mass of ammonia is needed to form 100 g of ammonium sulfate?

Step 1:

$M(NH_3) = 17.0\,g\,mol^{-1}$ and $M((NH_4)_2SO_4) = 132.1\,g\,mol^{-1}$

Step 2:

$$n((NH_4)_2SO_4) = \frac{100}{132} = 0.757\,mol$$

Step 3:

$n(NH_3) = 2 \times 0.757 = 1.51\,mol$
(note the 2:1 ratio in the equation)

Step 4:

$m(NH_3) = n \times M = 1.51 \times 17.0 = 25.7\,g$

WORKING OUT FORMULAE AND EQUATIONS FROM REACTING MASSES

You might assume that the formulae and equations for all reactions are already known. However, there are sometimes two or more possible formulae for a substance. There can also be more than one reaction for the same reactants. Reacting masses can be used to identify the correct formula, or which of the reactions is occurring.

WORKED EXAMPLE 3

Sodium carbonate exists as the pure (anhydrous) compound but also as three **hydrates**. Careful heating can decompose these hydrates to one of the other hydrates or to the anhydrous compound. The measurement of reacting masses can allow you to determine the correct equation for the decomposition.

Question

A 16.7 g sample of a hydrate of sodium carbonate ($Na_2CO_3.10H_2O$) is heated at a constant temperature for a specified time until the reaction is complete. A mass of 3.15 g of water is obtained. What is the equation for the reaction occurring?

Method

Step 1: calculate the molar masses of the relevant substances

$M(Na_2CO_3.10H_2O) = 286.1\,g\,mol^{-1}$
and $M(H_2O) = 18.0\,g\,mol^{-1}$

Step 2: calculate the amounts of these substances

$$Na_2CO_3.10H_2O\text{: } n = \frac{m}{M} = \frac{16.7}{286.1} = 0.0584\,mol$$

$$\text{water: } n = \frac{m}{M} = \frac{3.15}{18.0} = 0.175\,mol$$

Step 3: use these amounts to calculate the simplest whole-number ratio for these substances

$Na_2CO_3.10H_2O$ and H_2O are in the ratio 0.0584 : 0.175 or 1 : 3

Step 4: use this ratio to work out the equation for the reaction

$$Na_2CO_3.10H_2O \rightarrow Na_2CO_3.7H_2O + 3H_2O$$

We have worked out the $Na_2CO_3.7H_2O$ formula by considering the ratio of the other two formulae.

WORKED EXAMPLE 4

Copper forms two oxides. Both oxides can be converted to copper by heating with hydrogen.

Question

An oxide of copper is heated in a stream of hydrogen to constant mass. The masses of copper and water formed are Cu = 17.6 g and H_2O = 2.56 g. What is the equation for the reaction occurring?

Method

Step 1: $M(Cu) = 63.5\,g\,mol^{-1}$ and $M(H_2O) = 18.0\,g\,mol^{-1}$

Step 2: $n(Cu) = \frac{17.6}{63.5} = 0.277\,mol$ and $n(H_2O) = \frac{2.56}{18.0} = 0.142\,mol$

Step 3: ratio is 0.277 : 0.142 = 2 : 1

Step 4: the equation has 2 mol of Cu and 1 mol of H_2O, so the products must be $2Cu + H_2O$

So the equation is:

$$Cu_2O + H_2 \rightarrow 2Cu + H_2O \text{ and not } CuO + H_2 \rightarrow Cu + H_2O$$

LEARNING TIP

One important part of both these calculation methods is the use of the relevant ratio from the equation. Practise deciding which substances should be used for the ratio and which way round to use the ratio.

CHECKPOINT

SKILLS ADAPTIVE LEARNING

1. A fertiliser manufacturer makes a batch of 20 kg of ammonium nitrate. What mass of ammonia, in kg, does the manufacturer need to start with?
2. A sample of an oxide of iron was reduced to iron by heating with hydrogen. The mass of iron obtained was 4.35 g and the mass of water was 1.86 g. Deduce the equation for the reaction that occurred.



▲ **fig A** Chemistry on this scale needs careful calculations so that no reactants are wasted.

SUBJECT VOCABULARY

hydrate compound containing water of crystallisation, represented by formulae such as $CuSO_4.5H_2O$.

1C 4 THE YIELD OF A REACTION

SPECIFICATION REFERENCE 1.9

LEARNING OBJECTIVES

- Calculate percentage yields in laboratory and industrial processes using chemical equations and experimental results.



▲ **fig A** Pharmaceutical companies are always looking for ways to increase the percentage yield when manufacturing a drug.

THEORETICAL YIELD, ACTUAL YIELD AND PERCENTAGE YIELD

In the laboratory, when you are making a product, you naturally want to obtain as much of it as possible from the reactants you start with. In industry, where reactions occur on a much larger scale, and there is economic competition between manufacturers, it is even more important to maximise the product of a reaction.

There are some reasons why the mass of a reaction product may be less than the maximum possible.

- The reaction is reversible and so may not be complete.
- There are side-reactions that lead to other products that are not wanted.
- The product may need to be purified, which may result in loss of product.

TERMINOLOGY RELATING TO 'YIELD'

We normally use the term 'yield' with other words, such as:

- **theoretical yield**
- **actual yield**
- **percentage yield**.

In the laboratory, theoretical yield and actual yield may be measured in grams, but in industry, kilograms and tonnes are more likely to be used. Percentage yield is the term most often used, but you need to understand the other two terms first.

THEORETICAL YIELD

We calculate theoretical yield using the equation for the reaction, and we use a method you are familiar with from previous topics. It is always assumed that the reaction goes to completion, with no losses.

WORKED EXAMPLE 1

Copper(II) carbonate is decomposed to obtain copper(II) oxide. The equation for the reaction is:

$$CuCO_3 \rightarrow CuO + CO_2$$

What is the theoretical yield of copper(II) oxide obtainable from 5.78 g of copper(II) carbonate?

Step 1: calculate the amount of starting material

$$n(CuCO_3) = \frac{5.78}{123.5} = 0.0468 \text{ mol}$$

Step 2: use the reacting ratio to calculate the amount of desired product

$$n(CuO) = 0.0468 \text{ mol}$$

Step 3: calculate the mass of desired product

$$m = 0.0468 \times 79.5 = 3.72 \text{ g}$$

WORKED EXAMPLE 2

Magnesium phosphate can be prepared from magnesium by reacting it with phosphoric acid. The equation for the reaction is:

$$3Mg + 2H_3PO_4 \rightarrow Mg_3(PO_4)_2 + 3H_2$$

What is the theoretical yield of magnesium phosphate obtainable from 5.62 g of magnesium?

Step 1: $n(Mg) = \frac{5.62}{24.3} = 0.231 \text{ mol}$

Step 2: $n(Mg_3(PO_4)_2) = \frac{0.231}{3} = 0.0770 \text{ mol}$

Step 3: $m = 0.0770 \times 262.9 = 20.2 \text{ g}$

ACTUAL YIELD

This is the actual mass obtained by weighing the product obtained, not by calculation.

PERCENTAGE YIELD

Percentage yield is calculated using the equation:

$$\frac{\text{actual yield} \times 100}{\text{theoretical yield}} = \text{percentage yield}$$

This calculation may be done independently or in conjunction with the calculation of theoretical yield.

EXAM HINT

You may be asked in a question to suggest why you have a low yield. If so, you should give a specific example in the reaction method where you may have lost product such as, 'some solution was left on the filter paper during filtration'.

WORKED EXAMPLE 3

The theoretical yield in a reaction is 26.7 tonnes. The actual yield is 18.5 tonnes. What is the percentage yield?

$$\text{percentage yield} = \frac{18.5 \times 100}{26.7} = 69.3\%$$

WORKED EXAMPLE 4

A manufacturer uses this reaction to obtain methanol from carbon monoxide and hydrogen:

$$CO + 2H_2 \rightarrow CH_3OH$$

The manufacturer obtains 4.07 tonnes of methanol starting from 4.32 tonnes of carbon monoxide. What is the percentage yield?

First, calculate the theoretical yield.

Step 1: $n(CO) = \frac{4.32 \times 10^6}{28.0} = 1.54 \times 10^5$ mol

Step 2: $n(CH_3OH) = 1.54 \times 10^5$ mol (because of 1:1 ratio)

Step 3: $m = 1.54 \times 10^5 \times 32.0 = 4.94 \times 10^6$ mol

Then use that answer to calculate the percentage yield.

$$\text{Percentage yield} = \frac{4.07 \times 10^6 \times 100}{4.94 \times 10^6} = 82.4\%$$

LEARNING TIP

For calculations using kilograms and tonnes, remember that:

1 kg = 1×10^3 g

1 tonne = 1×10^6 g

CHECKPOINT

SKILLS | ADAPTIVE LEARNING

1. A student prepares a sample of copper(II) sulfate crystals, $CuSO_4.5H_2O$, weighing 7.85 g. She started with 4.68 g of copper(II) oxide. What is the percentage yield?

2. A manufacturer makes some ethanoic acid using this reaction:

$$CH_3OH + CO \rightarrow CH_3COOH$$

Starting with 50.0 kg of methanol, the manufacturer obtains 89.2 kg of ethanoic acid. What is the percentage yield?

SUBJECT VOCABULARY

theoretical yield the maximum possible mass of a product in a reaction, assuming complete reaction and no losses

actual yield the actual mass obtained in a reaction

percentage yield the actual yield divided by the theoretical yield, expressed as a percentage

1C 5 ATOM ECONOMY

SPECIFICATION REFERENCE 1.9

LEARNING OBJECTIVES

- Calculate percentage atom economies using chemical equations and experimental results.

BACKGROUND TO ATOM ECONOMY

In **Topic 1C.4**, we looked at the percentage yield of a reaction. The closer the value is to 100%, the better. A higher percentage means that less of the starting materials are lost or end up as unwanted products.

Percentage yield is an important factor to take into consideration when assessing the suitability of an industrial process. However, it is not the only one. Other factors include:

- the availability or scarcity of non-renewable raw materials
- the cost of raw materials
- the quantity of energy needed.

HOW ATOM ECONOMY WORKS

Here is an example of **atom economy** in action.

There are two main processes in the manufacture of phosphoric acid. To make the comparison easier to follow, a single summary equation is shown for each process.

Process 1 $Ca_3(PO_4)_2 + 3H_2SO_4 \rightarrow 2H_3PO_4 + 3CaSO_4$

Process 2 $P_4 + 5O_2 + 6H_2O \rightarrow 4H_3PO_4$

There are advantages and disadvantages of both processes. However, what you can see from these equations is that all of the atoms in the starting materials for Process 2 end up in the desired product. In Process 1, many of the atoms end up in a second, unwanted product, calcium sulfate. Process 1 has a lower atom economy than Process 2.

THE CONTRIBUTION OF BARRY TROST

A chemist from the USA, Barry Trost, developed the idea of atom economy as an alternative way of assessing chemical reactions, especially in industrial processes. He believed that it was important to consider how many atoms from the reactants end up in the desired product.



▲ **fig A** Barry Trost is a pioneer of the concept of atom economy.

The expression we use to calculate atom economy (usually described as percentage atom economy) is:

$$\text{atom economy} = \frac{\text{molar mass of the desired product}}{\text{sum of the molar masses of all products}} \times 100$$

You can see that you do not need a calculator to work out the atom economy of Process 2. There is only one product, so it must be 100%.

$$\text{For Process 1, atom economy} = \frac{(98.0 \times 2) \times 100}{(98.0 \times 2) + (136.2 \times 3)} = 32.4\%$$

So, you can see that less than one-third of the mass of the starting materials ends up in the desired product, which does not look good if the $CaSO_4$ is a waste product that has to be disposed of. If the other product has a use, then the manufacturer can sell it, which would partly balance the low atom economy of the process. Even if the percentage yield of Process 1 was as high as 100%, the atom economy is still only 32.4%.

REACTION TYPES AND ATOM ECONOMY

We can make some generalisations about certain types of reaction.

- Addition reactions have 100% atom economy.
- Elimination and substitution reactions have lower atom economies.
- Multistep reactions may have even lower atom economies.

EXAMPLES OF CALCULATIONS

WORKED EXAMPLE 1

Sodium carbonate is an important industrial chemical manufactured by the Solvay process. The overall equation for the process is:

$$CaCO_3 + 2NaCl \rightarrow Na_2CO_3 + CaCl_2$$

A manufacturer starts with 75.0 kg of calcium carbonate and obtains 76.5 kg of sodium carbonate. Calculate the percentage yield and atom economy for this reaction.

M_r values are 100.1 for $CaCO_3$ and 106.0 for Na_2CO_3.

$$\text{Theoretical yield} = \frac{75.0 \times 106.0}{100.1} = 79.4\,\text{kg}$$

$$\text{Percentage yield} = \frac{76.5 \times 100}{79.4} = 96.3\%$$

$$\text{Atom economy} = \frac{106.0 \times 100}{106.0 + 111.1} = 48.8\%$$

WORKED EXAMPLE 2

Hydrazine (N_2H_4) can be used as a rocket fuel and is manufactured using this reaction:

$$2NH_3 + NaOCl \rightarrow N_2H_4 + NaCl + H_2O$$

What is the atom economy for this reaction?

You first need to work out the molar masses of the products. These are 32.0, 58.5 and 18.0.

$$\text{Atom economy} = \frac{32.0 \times 100}{32.0 + 58.5 + 18.0} = 29.5\%$$

WORKED EXAMPLE 3

A manufacturer of ethene wants to convert some ethene into 1,2-dichloroethane. He considers two possible reactions:

Reaction 1 $H_2C{=}CH_2 + Cl_2 \rightarrow ClCH_2CH_2Cl$

Reaction 2 $2H_2C{=}CH_2 + 4HCl + O_2 \rightarrow 2ClCH_2CH_2Cl + 2H_2O$

Explain, without doing a calculation, which reaction would be a good choice on the basis of atom economy.

The answer is Reaction 1, because there is only one product, so all the atoms in the reactants end up in the desired product and the atom economy is 100%.

Reaction 2 has a lower atom economy because some of the atoms in the reactants form water, which has no value as a product.

CHECKPOINT

1. Ethanol can be manufactured by the hydration of ethene:

$$C_2H_4 + H_2O \rightarrow C_2H_5OH$$

What is the atom economy of this process?

2. Ethene can be manufactured by the dehydration of ethanol:

$$C_2H_5OH \rightarrow C_2H_4 + H_2O$$

What is the atom economy of this process?

SUBJECT VOCABULARY

atom economy the molar mass of the desired product divided by the sum of the molar masses of all the products, expressed as a percentage

LEARNING TIP

Remember that percentage yield indicates how efficient a reaction is at converting the reactants to the products. Atom economy indicates the percentage of atoms from the starting materials that end up in the desired product.

EXAM HINT

Also note that if you have more than one product, you will need time and energy to separate the desired product from the mixture.